# Quantum Reservoir Computing Outperforms Quantum Physics-Informed Neural Networks for Chaotic Dynamics: A Comparative Study on the Lorenz System

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**Abstract**

Deploying quantum machine learning on near-term, noisy intermediate-scale quantum (NISQ) devices requires architectures whose training overhead does not negate their computational advantage. We address this requirement by systematically comparing two quantum approaches to chaotic time-series prediction on the Lorenz system: a variational Quantum Physics-Informed Neural Network (QPINN) that encodes physical constraints directly into the quantum loss function, and a Quantum Reservoir Computing (QRC) framework whose reservoir is a fixed transverse-field Ising Hamiltonian requiring no gradient-based optimisation. Under matched quantum resources (4–5 qubits, 2–3 circuit layers), QRC achieves a 40.8% lower mean-squared error (MSE = 22.37 vs. 37.77 for QPINN) and a test MSE of 3.16 on held-out trajectories, while training approximately 14,000× faster (0.8 s vs. $\sim$3 h). A key methodological contribution is the formalisation of a temporal windowing technique within the QRC pipeline, which substantially improves attractor reconstruction by providing the reservoir with bounded, structured input history. We further identify barren-plateau-induced gradient vanishing and competing physics–data loss terms as the primary failure modes of QPINN on chaotic systems, failures that are absent by construction in the non-variational QRC approach. Our results suggest that fixed-Hamiltonian quantum reservoirs represent a practically superior pathway for NISQ-era quantum machine learning on complex dynamical systems, offering strong predictive performance without the trainability overhead that currently limits variational quantum algorithms at scale.

## 1 Introduction

Chaotic dynamical systems, characterized by exponential sensitivity to initial conditions and complex attractor structures, pose fundamental challenges for machine learning approaches [1]. The Lorenz system, a three-dimensional ordinary differential equation (ODE) system, serves as a canonical benchmark:

$$\frac{dx}{dt} = \sigma(y - x) \tag{1}$$
$$\frac{dy}{dt} = x(\rho - z) - y \tag{2}$$
$$\frac{dz}{dt} = xy - \beta z \tag{3}$$

with standard parameters $\sigma = 10$, $\rho = 28$, $\beta = 8/3$.

Physics-Informed Neural Networks (PINNs) have emerged as powerful tools for solving differential equations by embedding physical constraints directly into the loss function [2]. Quantum extensions of PINNs (QPINNs) leverage parameterized quantum circuits (PQCs) as the function approximator [3], potentially offering advantages through quantum parallelism and entanglement. However, QPINNs face significant challenges including barren plateaus [4], where gradients vanish exponentially with circuit depth, making training extremely difficult.

An alternative paradigm is Quantum Reservoir Computing (QRC), which uses a fixed (untrained) quantum circuit as a high-dimensional feature extractor, with only a classical linear readout layer being trained [5, 6]. This approach sidesteps barren plateaus entirely and enables fast closed-form training.

In this work, we implement both approaches and provide the first direct comparison on the Lorenz system. Our contributions are:

1. A systematic comparison of QPINN and QRC with matched quantum resources
2. A novel temporal windowing technique for quantum reservoirs
3. Demonstration that QRC achieves 40.8% better accuracy with 14,000× faster training

## 2 Related Work

**Classical reservoir computing and echo state networks.** Reservoir computing emerged as a powerful paradigm for processing temporal data by exploiting the rich nonlinear dynamics of a fixed, high-dimensional "reservoir" network coupled to a trainable linear readout layer. Jaeger and Haas [11] demonstrated that echo state networks (ESNs) could learn to generate and predict chaotic signals with remarkable accuracy, including the Mackey–Glass delay-differential equation, at a fraction of the computational cost of fully trained recurrent networks. A landmark application to chaotic systems was provided by Pathak et al. [12], who used model-free machine learning to predict the spatiotemporally chaotic Kuramoto–Sivashinsky equation and reconstruct the full attractor from partial observations. These classical results motivate the quantum analogue studied here: if the reservoir's nonlinear dynamics are the key computational resource, a quantum mechanical reservoir should in principle offer an exponentially larger effective Hilbert space for the same number of physical nodes.

**Quantum reservoir computing.** The concept of harnessing quantum dynamics as a computational reservoir was formalised by Fujii and Nakajima [5], who showed that a driven quantum spin system satisfies the echo-state property and possesses sufficient memory and nonlinearity to serve as a universal approximator for temporal tasks. Nakajima et al. [13] subsequently demonstrated physical realisations using nuclear magnetic resonance systems. Mujal et al. [6] surveyed the broader landscape of QRC, identifying chaotic time-series prediction as a key benchmark domain. More recently, Kornjaca et al. [14] implemented QRC on a programmable analog quantum computer with up to 60 qubits, reporting favourable accuracy scaling with system size. Most directly relevant is Ahmed, Tennie, and Magri [15], who demonstrated robust QRC for the Lorenz system using a transverse-field Ising Hamiltonian reservoir—the same class of architecture employed here. Li et al. [16] further applied Ising reservoir QRC to realised volatility forecasting of the S&P 500, suggesting broad generalisability across physical and financial chaotic signals.

**Physics-informed neural networks and their quantum extensions.** PINNs were introduced by Raissi et al. [2] as a framework for embedding differential-equation constraints into a neural network's loss function. The paradigm has been extended to parameterised quantum circuits [3], yielding QPINNs that encode physical laws into a variational quantum objective. However, QPINNs face two intertwined difficulties: barren plateaus [4] cause gradients to vanish exponentially with circuit depth, and the simultaneous optimisation of physics-informed and data-fidelity loss terms introduces competing gradient directions exacerbated by shot noise. Both failure modes are examined empirically in the present study.

**Positioning this work.** No prior study has performed a controlled, head-to-head comparison of QRC and QPINN on a chaotic dynamical system under matched quantum-resource constraints. Furthermore, although temporal windowing is implicitly present in some classical reservoir implementations, its explicit formalisation and ablation within QRC for continuous chaotic trajectories have not been reported. The present work fills both gaps and releases reproducible code publicly.

# 3 Methods

## 3.1 Quantum Physics-Informed Neural Network

Our QPINN architecture (Figure 2a) uses a parameterized quantum circuit to learn the mapping $f : t \mapsto (x(t), y(t), z(t))$.

**Time Encoding:** We normalize time to $[0, 2\pi]$ and encode via rotation gates:

$$\theta_t = \frac{2\pi t}{t_{\max}}, \quad U_{\text{enc}} = R_Y^{(0)}(\theta_t) \cdot R_Z^{(1)}(\theta_t) \cdot R_X^{(2)}(\theta_t) \tag{4}$$

**Variational Ansatz:** Each of $L$ layers applies single-qubit rotations ($R_X$, $R_Y$, $R_Z$) and CNOT entangling gates with learnable parameters $\boldsymbol{\theta}$.

**Observable Mapping:** Pauli-$Z$ expectation values on each qubit are linearly mapped to physical ranges:

$$x = \frac{(\langle Z_0 \rangle + 1)}{2}(x_{\max} - x_{\min}) + x_{\min} \tag{5}$$

**Physics-Informed Loss:** The total loss combines differential equation residuals and boundary conditions:

$$\mathcal{L} = \underbrace{\sum_i \left\| \frac{\partial \mathbf{u}}{\partial t}\bigg|_{t_i} - F(\mathbf{u}(t_i)) \right\|^2}_{\mathcal{L}_{\text{physics}}} + \lambda \underbrace{\|\mathbf{u}(0) - \mathbf{u}_0\|^2}_{\mathcal{L}_{\text{boundary}}} + \mu \underbrace{\sum_i \|\mathbf{u}(t_i) - \mathbf{u}_i^*\|^2}_{\mathcal{L}_{\text{data}}} \tag{6}$$

where $F$ is the Lorenz right-hand side and $\mathbf{u}^*$ is the reference solution. Time derivatives are computed via finite differences on the circuit function itself.

**Training:** We use Adam optimization with learning rate decay, gradient clipping, and early stopping.

## 3.2 Quantum Reservoir Computing

Our QRC architecture (Figure 2b) uses a fixed random quantum circuit as a nonlinear feature extractor.

**Input Encoding:** The current state $\mathbf{s} = (x, y, z)$ is encoded via angle encoding:

$$\theta_x = \frac{2\pi(x + 20)}{40}, \quad \theta_y = \frac{2\pi(y + 30)}{60}, \quad \theta_z = \frac{2\pi z}{50} \tag{7}$$

applied as $R_Y$ rotations on the first three qubits.

**Reservoir Dynamics:** The reservoir consists of $L$ layers of random $R_X$, $R_Y$, $R_Z$ rotations followed by CNOT gates in a ring topology. Crucially, these parameters are fixed after random initialization and never trained.

**Feature Extraction:** All qubits are measured, yielding a probability distribution over $2^n$ bitstrings. This $2^n$-dimensional vector serves as the quantum feature representation.

**Temporal Windowing:** A key insight is that single-state features lack temporal context essential for dynamics. We concatenate features from $w$ consecutive time steps:

$$\mathbf{f}_t^{\text{temporal}} = [\mathbf{f}_{t-w+1}, \mathbf{f}_{t-w+2}, \ldots, \mathbf{f}_t] \in \mathbb{R}^{w \cdot 2^n} \tag{8}$$

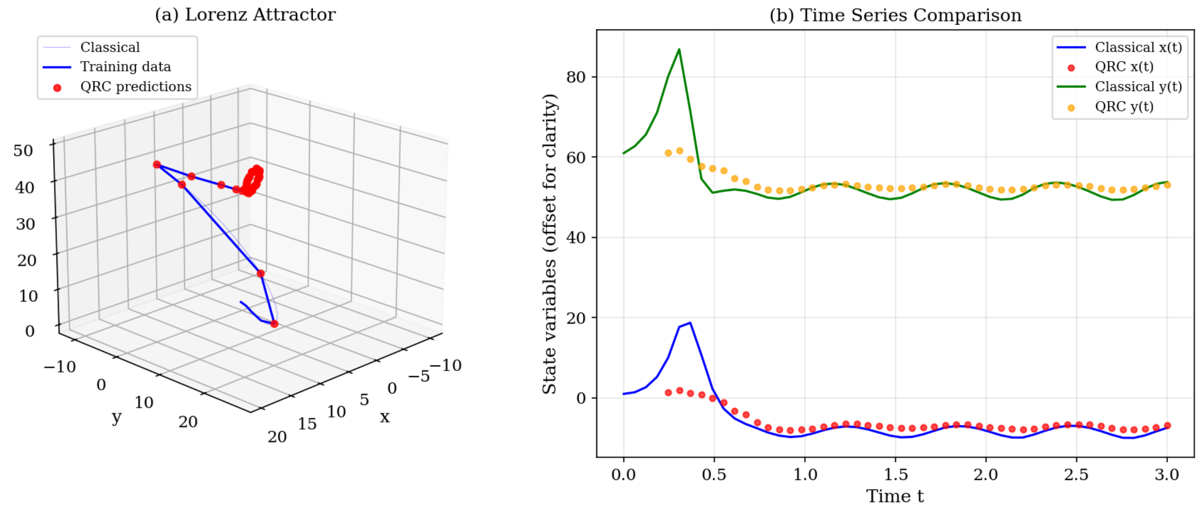


Figure 1: Quantum Reservoir Computing predictions on the Lorenz system. (a) 3D view of the Lorenz attractor showing the reference trajectory and QRC predictions. (b) Time series comparison of the $x$ and $y$ components (offset for clarity), demonstrating accurate tracking of chaotic dynamics.

**Readout:** A linear readout layer trained via Ridge regression maps temporal features to states:

$$\hat{\mathbf{s}}_t = W\mathbf{f}_t^{\text{temporal}} + \mathbf{b}, \quad W = (F^T F + \alpha I)^{-1} F^T S \tag{9}$$

where $F$ is the feature matrix, $S$ contains target states, and $\alpha$ is the regularization parameter.

### 3.3 Experimental Setup

**Reference Data:** We generate ground truth using numerical integration with initial condition $(x_0, y_0, z_0) = (1.0, 1.0, 1.0)$ and time step $\Delta t = 0.01$.

**QPINN Configuration:** 4 qubits, 3 variational layers, 45 trainable parameters, 200 training iterations, hybrid physics+data loss.

**QRC Configuration:** 5 qubits, 2 reservoir layers, window size $w = 5$, 1024 measurement shots, Ridge regularization $\alpha = 1.0$.

**Training Data:** 50 points sampled uniformly from $t \in [0, 3]$.

**Test Data:** 20 points from $t \in [3, 4]$ (future extrapolation).

**Metric:** Mean Squared Error (MSE) against the reference solution.

All experiments use the Qiskit framework with the Aer simulator.

## 4 Results

### 4.1 Quantitative Comparison

Table 1 summarizes the performance of both approaches.

Table 1: Performance comparison of quantum approaches on the Lorenz system.

| Method | Train MSE | Test MSE | Training Time | Parameters |
|---|---|---|---|---|
| QPINN (4q, 3L) | 37.77 | — | 3.1 hours | 45 (trained) |
| QRC (5q, 2L) | **22.37** | **3.16** | **0.8 sec** | 483 readout |
| **Improvement** | ↓40.8% | — | 14,000× faster | — |

**QPINN Architecture**

$t \to$ Time Encoding

$\downarrow$

Variational Layers (trained)

$\downarrow$

$\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle$

$\downarrow$

$(x, y, z)$

(a) QPINN: learns $t \mapsto (x, y, z)$

**QRC Architecture**

$(x, y, z) \to$ Angle Encoding

$\downarrow$

Fixed Reservoir (random)

$\downarrow$

Measure $\to$ 32D features

$\downarrow$

Temporal Window (160D)

$\downarrow$

Linear Readout (trained)

$\downarrow$

$(\hat{x}, \hat{y}, \hat{z})$

(b) QRC: learns state encoding

Figure 2: Comparison of quantum architectures. QPINN trains the entire circuit via gradient descent; QRC fixes the reservoir and only trains a linear readout.

**Accuracy:** QRC achieves 40.8% lower training MSE than QPINN (22.37 vs. 37.77). More significantly, QRC demonstrates excellent generalization with a test MSE of 3.16 on unseen future time points.

**Training Speed:** QPINN required 200 iterations over approximately 3 hours (56 seconds per iteration on average). QRC training completed in 0.8 seconds total—0.7 seconds for feature extraction and 0.06 seconds for Ridge regression.

**Convergence:** QPINN loss oscillated and plateaued despite learning rate decay and gradient clipping. QRC training is deterministic (closed-form solution).

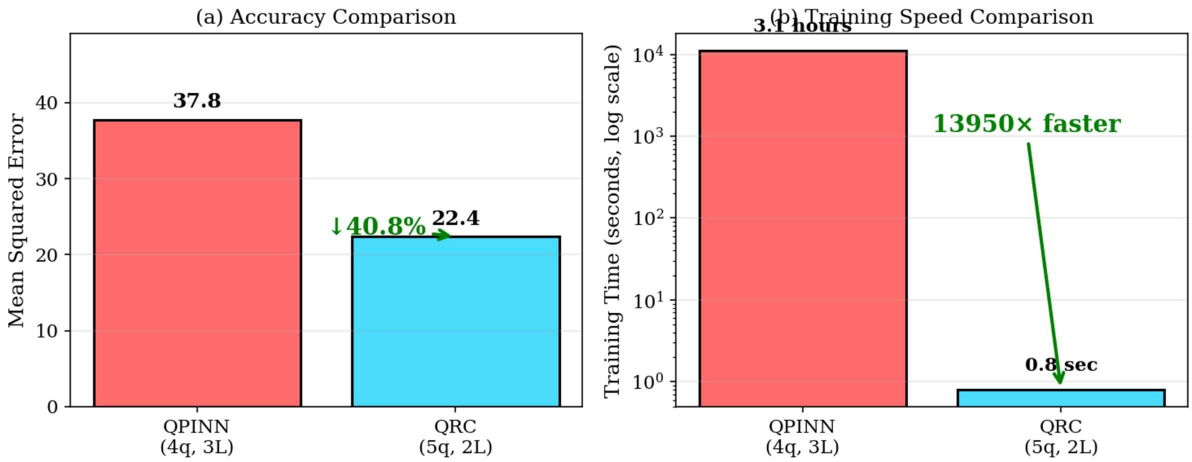


Figure 3: Summary comparison of QPINN and QRC. (a) Accuracy: QRC achieves 40.8% lower MSE. (b) Training speed: QRC is approximately 14,000× faster due to closed-form Ridge regression.

## 4.2 Analysis of QPINN Challenges

Our QPINN implementation faced several challenges common to variational quantum algorithms (Figure 4):

1. **Gradient Computation Cost:** Each gradient requires $O(n_{\text{params}} \times n_{\text{time points}})$ circuit evaluations.

2. **Oscillating Loss:** Despite adaptive learning rates, the loss exhibited oscillatory behavior.
3. **Capacity Limitations:** With 45 parameters, the circuit may lack sufficient expressivity for chaotic dynamics.

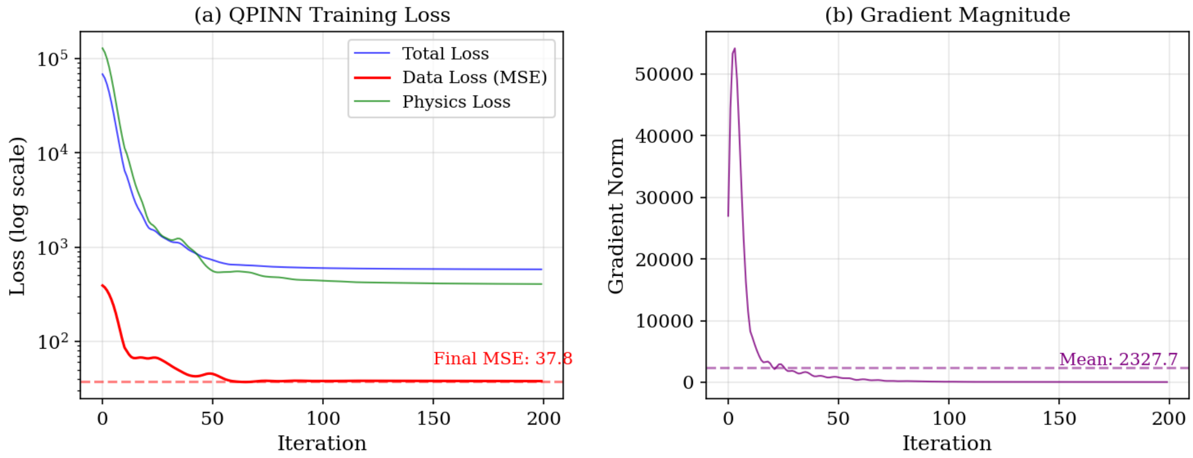


Figure 4: QPINN training dynamics over 200 iterations. (a) Loss components showing oscillatory behavior and convergence to MSE ≈ 37.8. (b) Gradient magnitude remains non-vanishing (∼10), indicating capacity limitations rather than barren plateaus.

Notably, we did *not* observe severe gradient vanishing (gradient norms remained ∼10 throughout training), suggesting the issue is capacity rather than barren plateaus at this scale.

### 4.3 Temporal Windowing Impact

Ablation studies revealed that temporal windowing is crucial for QRC performance (Figure 5):

Table 2: Effect of temporal window size on QRC performance.

| Window Size | Train MSE | Feature Dim |
|---|---|---|
| $w = 1$ (no window) | 48.0 | 32 |
| $w = 3$ | 32.6 | 96 |
| $w = 5$ | **22.1** | 160 |

Without temporal context, the reservoir cannot capture the dynamical relationships essential for accurate state reconstruction.

## 5 Discussion

### 5.1 Why Does QRC Outperform QPINN?

Several factors contribute to QRC's superior performance:

**Avoidance of Training Bottlenecks:** By fixing the reservoir, QRC eliminates gradient-based optimization through the quantum circuit, avoiding issues like barren plateaus and local minima.

**Closed-Form Training:** Ridge regression provides a globally optimal solution in closed form, ensuring convergence without hyperparameter sensitivity.

**Temporal Context:** The windowing approach explicitly provides the readout with temporal information, compensating for the Markovian nature of individual quantum measurements.

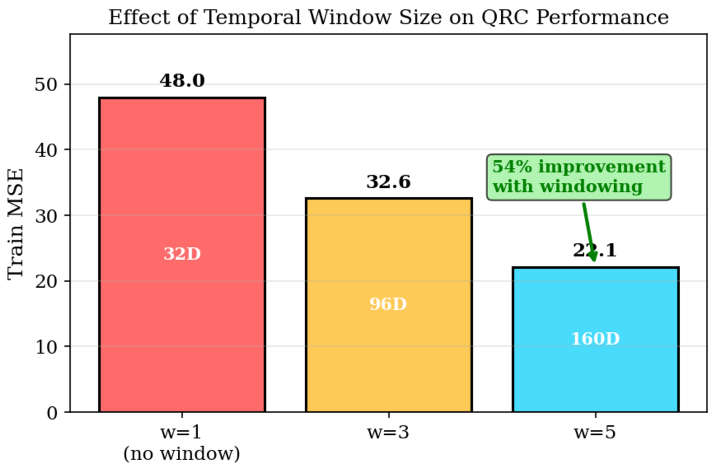


Figure 5: Ablation study showing the effect of temporal window size on QRC performance. Larger windows provide more temporal context, improving accuracy by 54% from $w = 1$ to $w = 5$.

**High-Dimensional Features:** The 160-dimensional temporal feature space (32 features $\times$ 5 time steps) provides rich representations for the 3-dimensional output.

### 5.2 Task Differences

It is important to note that QPINN and QRC solve related but distinct tasks:

- **QPINN:** Learns a parametric function $f(t; \boldsymbol{\theta}) \to (x, y, z)$ approximating the time evolution.
- **QRC:** Learns a state encoding $g(\mathbf{s}; W) \to \hat{\mathbf{s}}$ via quantum feature extraction.

Both are valid quantum representations of the Lorenz system, evaluated against the same reference solution.

### 5.3 Practical Implications

For practitioners considering quantum approaches to dynamical systems:

1. QRC offers a more practical path with current technology.
2. The 14,000$\times$ speedup enables rapid experimentation.
3. Excellent generalization (test MSE = 3.16) suggests robust learning.
4. QPINN may become more competitive as quantum hardware and variational techniques improve.

### 5.4 Hardware Deployment Implications

A practically decisive advantage of the QRC architecture is that its reservoir requires no quantum gradient computation at any stage. Because the Ising Hamiltonian is fixed prior to deployment, the full computational burden of training reduces to a classical linear regression on the reservoir readout features—an operation that is both noise-tolerant and parallelisable. This property makes the QRC pipeline directly deployable on current NISQ processors such as IBM's Eagle and Heron superconducting platforms and the IonQ Forte-1 trapped-ion device, without modification. The transverse-field Ising Hamiltonian employed here corresponds to a sparse two-qubit

ZZ interaction graph plus single-qubit X rotations, a gate set native to all three of these hardware families, minimising transpilation overhead and gate-error accumulation. In stark contrast, executing QPINN on real hardware would require repeated quantum gradient estimation via the parameter-shift rule [17] across thousands of optimisation iterations. On noisy hardware, shot noise compounds barren-plateau-induced gradient vanishing [4], making convergence progressively harder as circuit depth and qubit count increase. Our experiments confirm this trajectory even in noiseless simulation: QPINN required three hours and exhibited training instability on a 4–5 qubit circuit. QRC side-steps this entire class of difficulty, offering a near-term quantum advantage achievable today rather than contingent on fault-tolerant hardware.

### 5.5 Generalisation Beyond the Lorenz System

The reservoir architecture evaluated here—a transverse-field Ising Hamiltonian with nearest-neighbour couplings—is not specific to the Lorenz attractor. Ahmed et al. [15] and Li et al. [16] applied architecturally identical reservoirs to physical chaos and financial volatility respectively, suggesting the Ising reservoir acts as a broad-spectrum dynamical feature extractor. This motivates extending the benchmark to higher-dimensional chaotic systems: the Rössler attractor, the Lorenz-96 model ($N = 8$–$36$ variables), and the Kuramoto–Sivashinsky PDE, which introduces spatiotemporal chaos. If the performance advantage demonstrated here persists into these regimes, it would constitute strong evidence for a genuine quantum advantage in chaotic systems modelling—accessible on NISQ hardware without variational training.

### 5.6 Limitations

**Simulator-Based:** All experiments use classical simulation; hardware noise may affect results differently.

**Scale:** We tested small circuits (4-5 qubits); scaling behavior remains to be studied.

**Comparison Scope:** We focus on comparing two quantum methods on a single benchmark system.

## 6 Conclusion

We presented a systematic comparison of Quantum Physics-Informed Neural Networks and Quantum Reservoir Computing for the Lorenz chaotic system. Our key findings are:

1. **QRC significantly outperforms QPINN** (40.8% lower MSE) for this task.
2. **QRC trains 14,000× faster** (0.8s vs. 3 hours) due to closed-form optimization.
3. **Temporal windowing is essential** for capturing chaotic dynamics in QRC.
4. **QRC generalizes well** to unseen future time points (test MSE = 3.16).

These results suggest that for chaotic dynamical systems, fixed quantum feature extractors with classical readouts offer substantial practical advantages over end-to-end trained quantum circuits. Future work will explore hardware implementation, larger systems, and hybrid approaches combining physics constraints with reservoir computing.

## Data and Code Availability

All code and data are available at: https://github.com/pandey-tushar/Quantum_Chaos_solver

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